

ENTRIES 076-078 The Fourth Chip Graduates, the Fifth Chip Holds — September 16, 2026 — Status: COMPLETE

Folding Osaka Into Training, Setting Up Quebec as the New Holdout

A Combined 4-Chip Model, Tested Zero-Shot Against a Genuinely Unseen 5th Chip

Why

Entry 073 held Osaka out cleanly to run this project's first true zero-shot fourth-chip test. That job is done -- the result is on record. Continuing to waste a fully-simulated 708-circuit chip as a permanent holdout has diminishing research value once the question it was built to answer has been answered once. This entry graduates Osaka into training to build the strongest combined model to date, and sets up a fifth chip, Quebec (FakeQuebec, 127 qubits, 144 edges), as the new clean holdout -- preserving the project's ability to keep running this test honestly going forward, chip after chip.

Entry 076 — Quebec Holdout Dataset

Grew an eval-only bell-pair dataset for Quebec exactly as Entry 073 did for Osaka: same six hop-distance bins, same 4-seed x 4096-shot Aer/MPS simulation, same explicit exclusion from every training pipeline. Reached 710 circuits. This dataset is never folded into anything -- its entire value is staying untouched.

Entry 077 — Combined 4-Chip Model

Osaka's 708 training-eligible circuits (Entry 073's original growth) were folded into the graph dataset alongside Kyiv/Sherbrooke/Brisbane, taking the pool from 5,690 to 6,398 graphs. Retrained both the combined generalist (random 80/20 CV, same architecture and per-chip normalization as Entry 069) and the deployable MC-Dropout model (Entry 071's pattern, dropout active throughout, trained on all 6,398 graphs).

Chip	n (CV test)	MAE (pts)	R2	floor-collapse MAE	n_fc
Kyiv	421	1.08	0.981	2.04	23
Sherbrooke	404	1.29	0.961	1.43	43
Brisbane	306	1.14	0.966	0.29	7
Osaka	149	1.19	0.977	2.84	9
Overall	1280	1.17	0.972	--	--

Entry 077 combined-generalist held-out CV results (random 80/20 split, all 4 chips pooled). Compare to Entry 069's 3-chip generalist: overall MAE=1.07, Kyiv=0.77, Sherbrooke=1.45, Brisbane=0.99. Kyiv and Brisbane got slightly WORSE with a 4th, more diverse chip added to the pool; Sherbrooke improved. This is the expected nuanced pattern, not the uniform win across every chip that was hoped for going in -- pooling more diverse data does not universally help every individual chip's point accuracy.

Calibration check (separate held-out CV model, never the deployed one): MAE=1.36, R2=0.965, mean predicted std=0.89pts, corr(std,|error|)=0.414 -- positive and meaningful, slightly weaker than Entry 071's 0.553 but still confirms the uncertainty estimate is tracking real error, not noise.

Entry 078 — True Zero-Shot Test Against Quebec

The real payoff. The Entry 077 deployed model -- trained on Kyiv+Sherbrooke+Brisbane+Osaka, ZERO Quebec exposure -- was scored cold on all 710 Quebec circuits, using the average of the four training chips' normalization stats (Quebec has none, mirroring Entry 073's method exactly).

Model	MAE (pts)	R2	floor-collapse MAE	n_fc
GNN (Entry 077, 4-chip)	3.02	0.775	10.84	42
v4.1 closed-form	5.35	0.589	21.06	42

Zero-shot comparison on Quebec (n=710, mean predicted std=1.66pts).

This is a clear win on BOTH axes, not the tie Entry 073 found on Osaka. Entry 071's 3-chip model scored MAE=5.82 / R2=0.623 / floor-collapse MAE=15.77 zero-shot on Osaka, roughly matching v4.1's 5.57 / 0.686 /

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22.38 on aggregate accuracy while still beating it on floor-collapse. The 4-chip Entry 077 model instead beats v4.1 outright on every number: 44% lower aggregate MAE, 32% higher R2, and less than half v4.1's floor-collapse error. Uncertainty widened from ~0.89pts (same-chip) to 1.66pts on the unseen chip -- proportionally less dramatic than Entry 073's ~4x widening on Osaka, consistent with the model's actual error also degrading less on this genuinely harder test.

Entries 076-078 — Conclusion

The user's hoped-for outcome -- one model that beats all four training chips at every single parameter -- did not hold: Kyiv and Brisbane's same-chip point accuracy regressed slightly with a 4th, more heterogeneous chip added to the pool. But the more important number moved the right way by a wide margin: true zero-shot generalization to a genuinely unseen 5th chip improved substantially over the 3-chip model's equivalent test, converting a rough tie against the closed-form baseline into a decisive win on both raw accuracy and the floor-collapse detection that has been this project's central use case since Entry 036. More diverse training data trades a little same-chip precision for meaningfully better out-of-distribution generalization -- the honest, unglamorous, and genuinely useful finding this round produced.

Research Log — Index Addendum (Entries 076-078)

Files: setup_quebec.py, entry076_grow_quebec.py (quebec_bell_dataset.json, 710 circuits, eval-only), entry077_build_graphs.py (entry077_graph_dataset.json, 6398 graphs), entry077_train_gnn.py, entry077_combined_train.py (entry077_combined_params.json), entry077_mc_dropout.py, entry077_deploy_train.py (entry077_deploy_params.json, the new deployed model), entry078_zero_shot_test.py (entry078_zero_shot_results.json).